



# CJC

Complete Java Classes

by **Kunal Sir**

## Introduction Of Java:

Java is general purpose, class based, object-oriented programming language designed for lesser implementation dependencies. It is a computing platform for application development. Java is fast, secure, and reliable, therefore. It is widely used for developing Java applications in laptops, data centers, game consoles, scientific supercomputers, cell phones, etc.

**James Gosling** developed the Java platform at Sun Microsystems, and the Oracle Corporation later acquired it.

## History Of Java:

The history of Java is very interesting. Java was originally designed for interactive television, but it was too advanced technology for the digital cable television industry at the time.

The history of java starts from Green Team. Java team members (also known as Green Team), initiated this project to develop a language for digital devices such as set-top boxes, televisions etc. But it was suited for internet programming. Later, Java technology was incorporated by Netscape. The principles for creating Java programming were "Simple, Robust, Portable, Platform-independent, Secured, High Performance, Multithreaded, Architecture Neutral, Object-



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Oriented, Interpreted and Dynamic". Currently, Java is used in internet programming, mobile devices, games, e-business

solutions etc. There are given the major points that describes the history of java.

- I. James Gosling, Mike Sheridan, and Patrick Naughton initiated the java
- II. language project in June 1991.
- III. The small team of sun engineers called Green Team.
- IV. Originally designed for small, embedded systems in electronic appliances like set-top boxes.
- V. Firstly, it was called "Greentalk" by James Gosling and file extension was .gt. After that, it was called "Oak" and was developed as a part of the Greenproject.

### Features of Java:

#### 1) Simple: -

Java is very easy to learn, and its syntax is simple, clean and easy to understand. According to Sun Microsystem, Java language is a simple programming language because:

java syntax is based on C++ (so easier for programmers to learn it after C++).



## 2) Object Oriented: -

Java is an object-oriented programming language. Everything in java is an object. Object-oriented means we organize our software as a combination of different

types of objects that incorporate both data and behavior.

## 3) Platform Independent: -

First and foremost, the compiler converts the source code to its bytecode, and later on, JVM executes the bytecode generated by its compiler. Further, this bytecode can be run on any other platform, for example Windows, Linux and macOS.

## 4) Secured: -

Java is secured because

- No explicit pointer
- Java Programs run inside a virtual machine sandbox
- Class-loader: Class-loader in Java is a part of the Java Runtime Environment (JRE) which is used to load Java classes into the Java Virtual Machine dynamically. It adds security by separating the



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package for the classes of the local file system from those that are imported from network sources.

- **Bytecode Verifier:** It checks the code fragments for illegal code that can violate access rights to objects.
- **Security Manager:** It determines what resources a class can access such as reading and writing to the local disk.

## 5) **Robust:** -

It uses strong memory management.

There is a lack of pointers that avoids security problems.

Java provides automatic garbage collection which runs on the Java Virtual Machine to get rid of objects which are not being used by a Java application anymore.

There are exception handling and the type checking mechanism in Java. All these points make Java robust.



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## 6) Portable: -

Java is portable because it facilitates you to carry the Java bytecode to any platform. It doesn't require any implementation.

## Components of Java programming language:

There are three main components of Java language: **JVM**, **JRE**, and **JDK**.

**JVM** Stands for -- Java Virtual Machine.

**JRE** Stands for -- Java Runtime Environment.

**JDK** Stands for -- Java Development Kit.

## Java Virtual Machine: -

**Ever heard about WORA?**



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### Write once Run Anywhere

Well, Java applications are called WORA because of their ability to run a code on any platform. This is done only because of JVM. The JVM is a Java platform component that provides an environment for executing Java programs. JVM interprets the bytecode into machine code which is executed in the machine in which the Java program runs.

JVM performs the following functions:

- Loads the code
- Verifies the code
- Executes the code
- Provides runtime environment



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## Java Runtime Environment: -

The JRE software builds a runtime environment in which Java programs can be executed.

The JRE is the on-disk system that takes your Java code, combines it with the needed libraries, and starts the JVM to execute it.

The JRE contains libraries and software needed by your Java programs to run.

## Java Development Kit

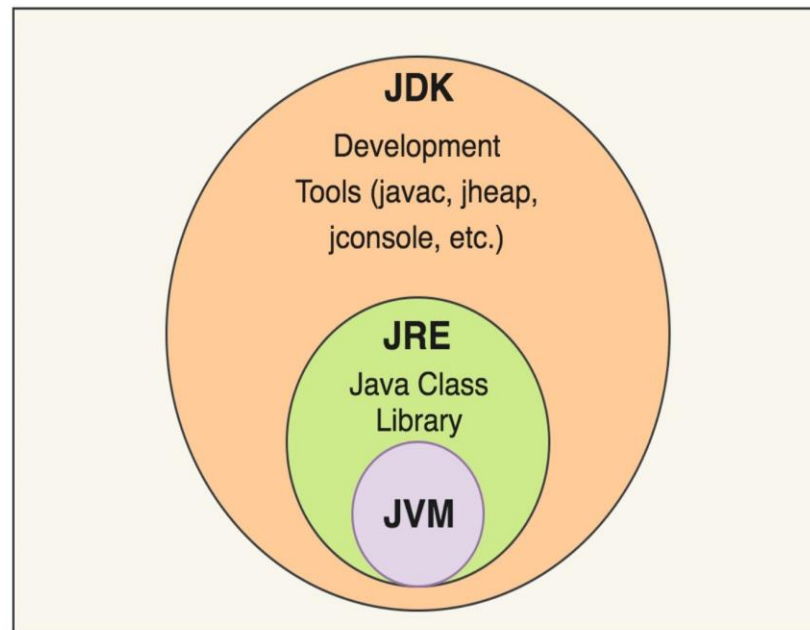
The Java Development Kit (JDK) is a software development environment used to develop Java applications and applets.

It contains JRE and several development tools, an interpreter/loader (java), a compiler (javac), an archiver (jar), a documentation generator (javadoc) accompanied with another tool.

## JVM, JRE and JDK Relation Diagram:



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## Step By Step Execution of Java Program:

Whenever, a program is written in Java, the javac compiles it.

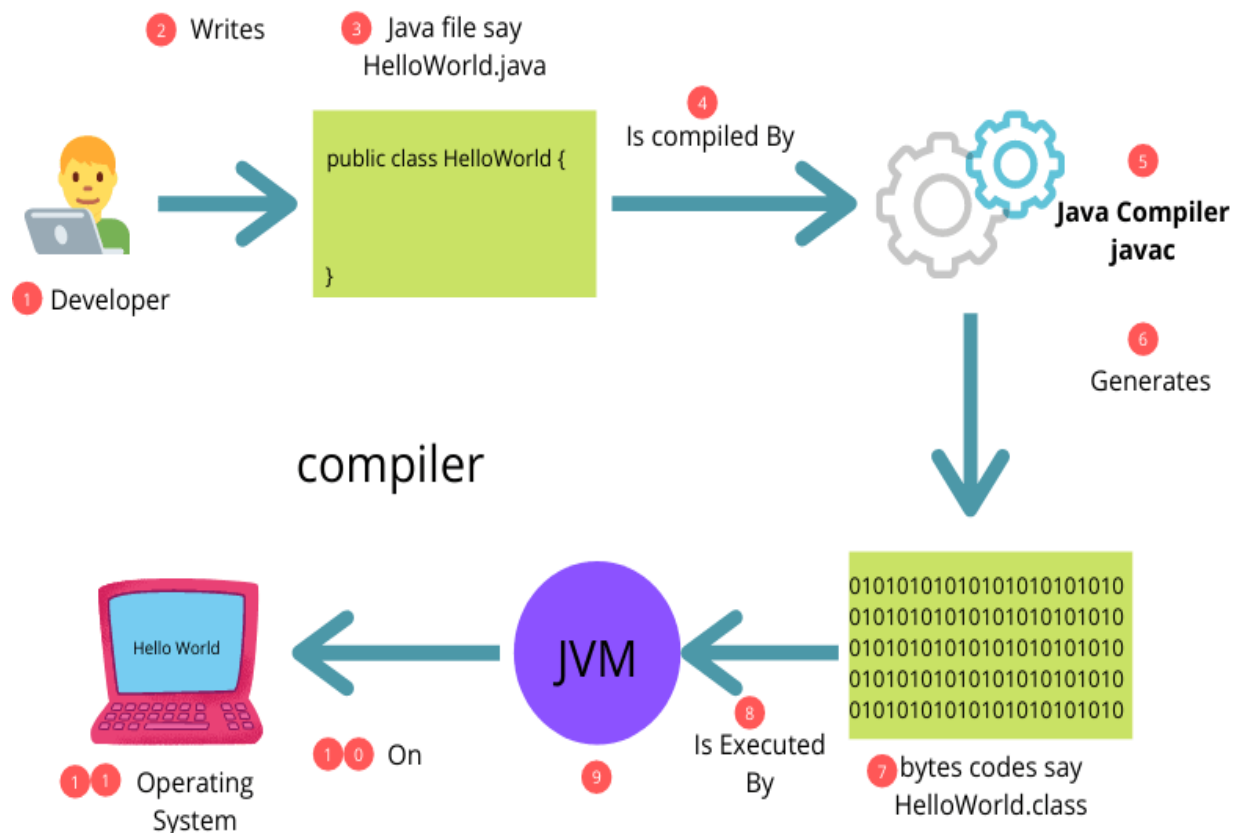
The result of the Java compiler is the .class file or the bytecode and not the machine native code (unlike C compiler).

The bytecode generated is a non-executable code and needs an interpreter to execute on a machine. This interpreter is the JVM and thus the bytecode is executed by the JVM.

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And finally, program runs to give the desired output.

In case of C or C++ (language that are not platform independent), the compiler generates an .exe file which is OS dependent. When we try to run this .exe file on another OS it does not run, since it is OS dependent and hence is not compatible with the other OS.



## Installation Of Jdk:

**Step 1:** Download the latest version of JDK from official website



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of Oracle or from the link

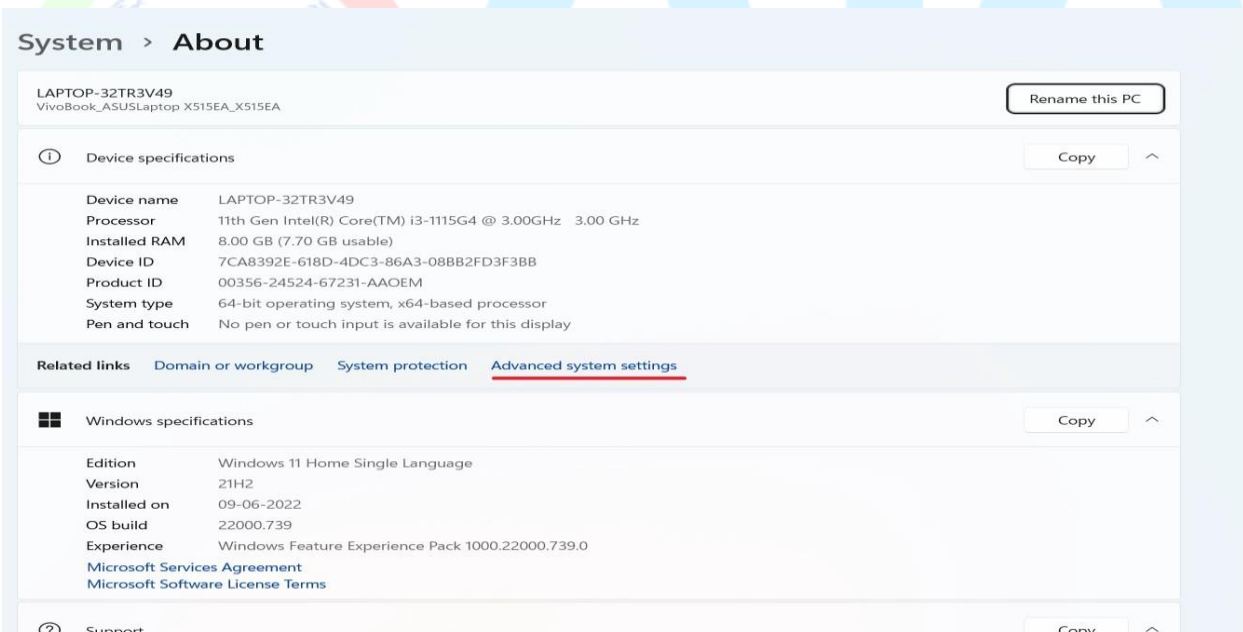
**Step 2:** After downloading the JDK, double click on the application file and click on Next. And finish the installation.

**Step 3:** Go to the installation folder location in C:\ drive, like below,

*i.e. C:\Program Files\Java\jdk1.8.0\bin*

**Step 4:** Right click on “**MyComputer**” and open **properties Settings**.

**Step 5:** click on Advance System Settings



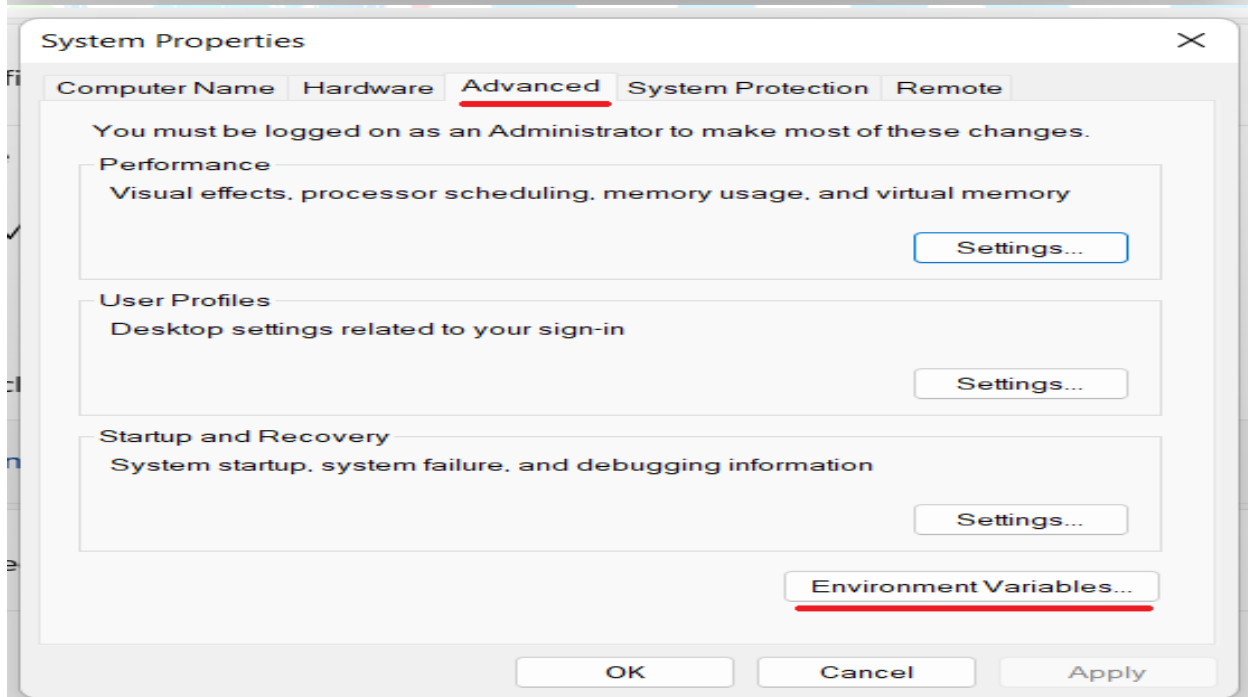
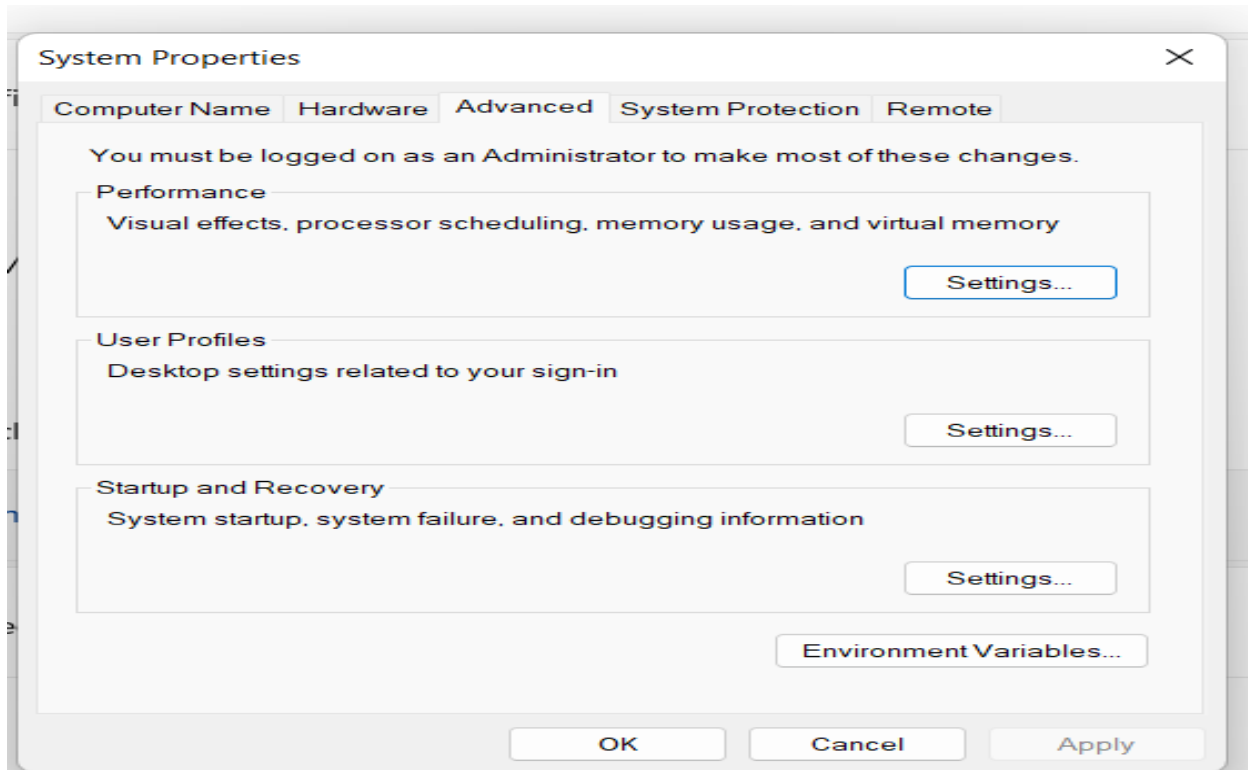
**Step 6:** Click on **Advanced** tabbed pane and after that click on **Environment Variables Button**.



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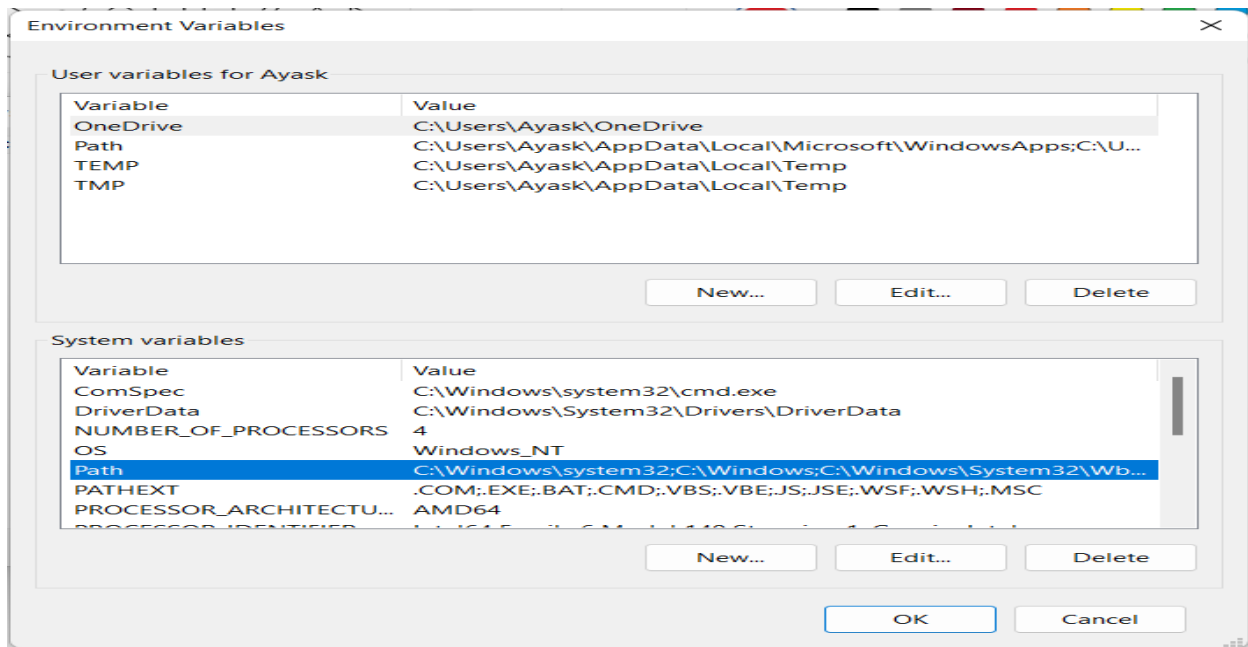


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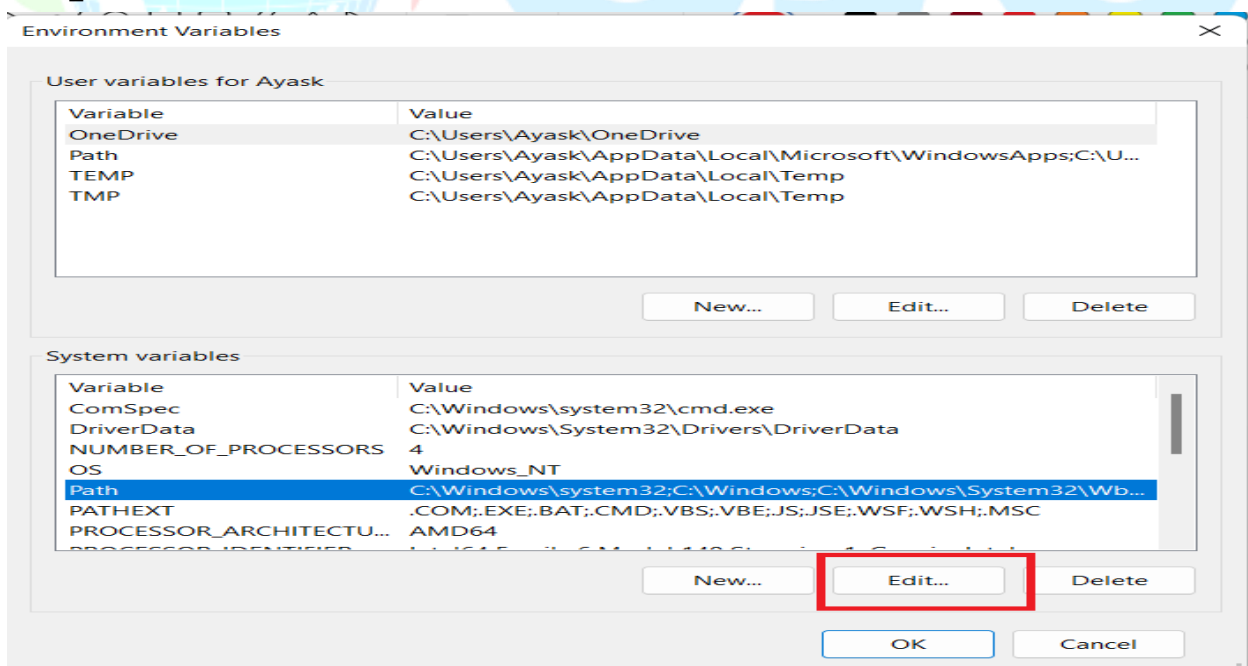
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**Step 7:** Select the **Path** variable in **System variables** dialog box-



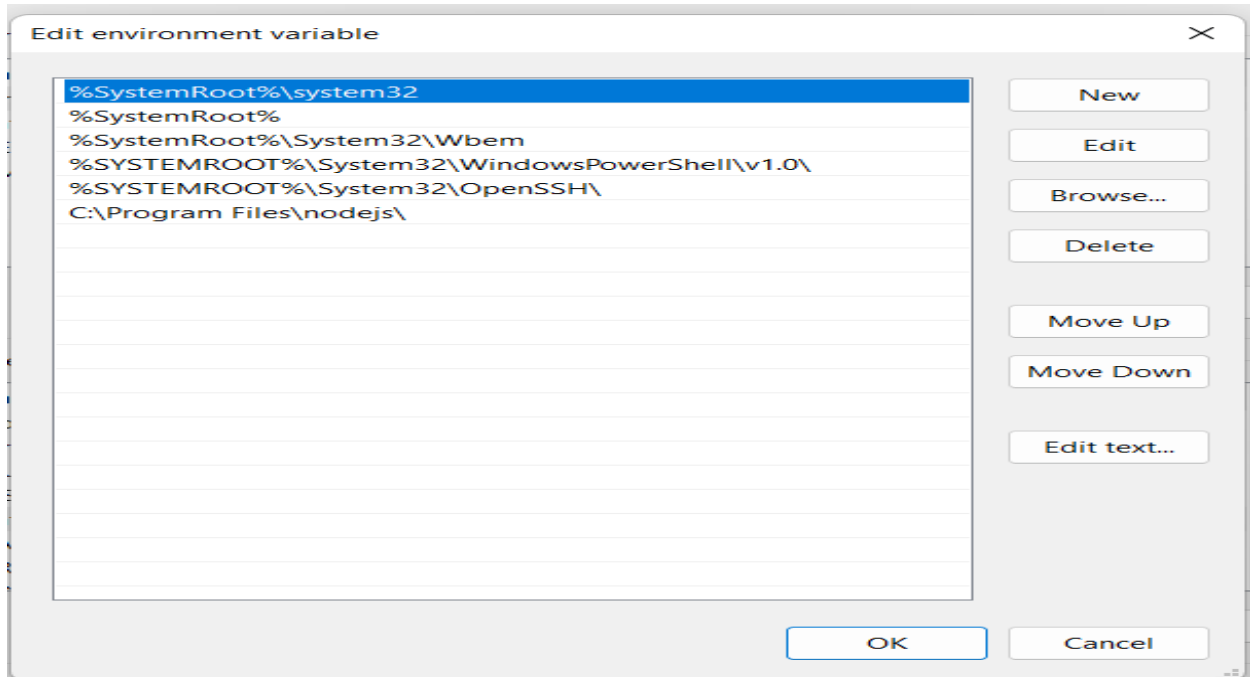
**Step 8:** click on edit button-



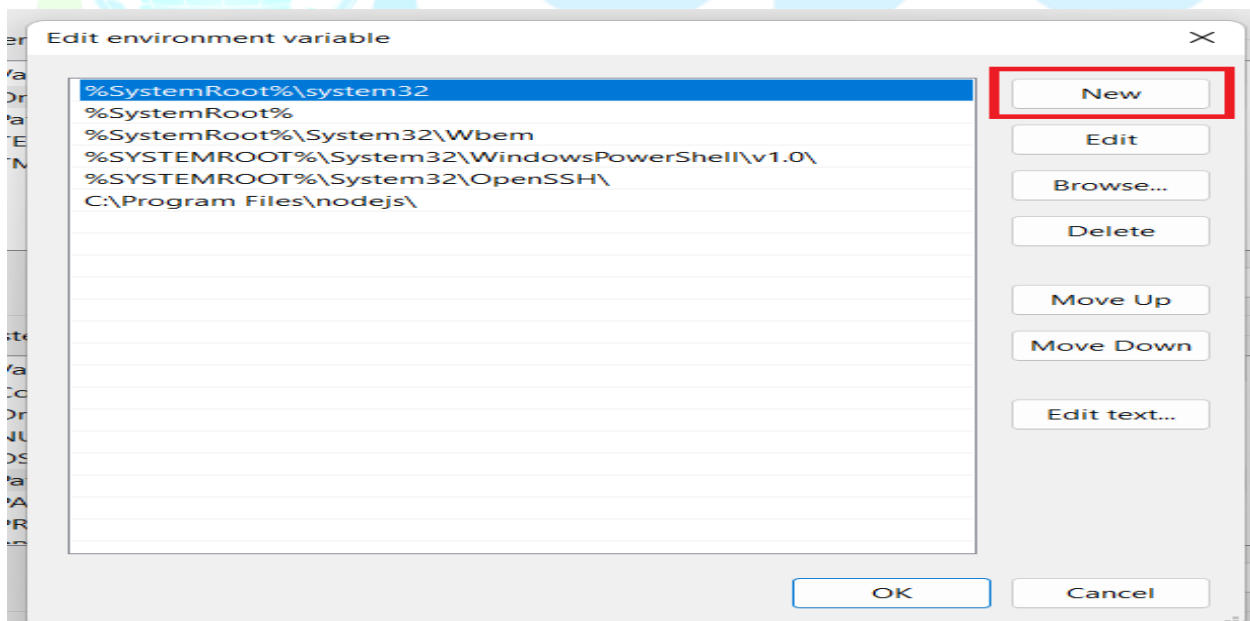


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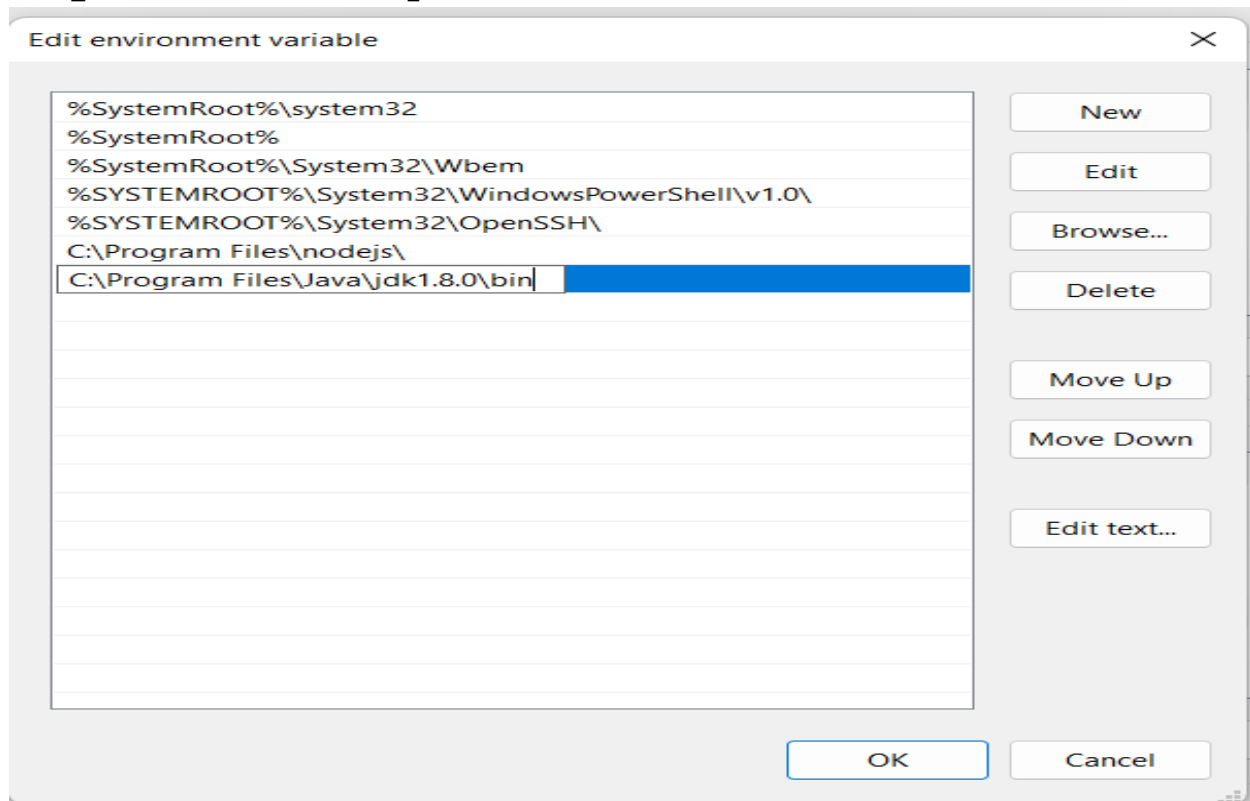
**Step 9:** next dialog box will be shown as below-



**Step 10:** Click on **New** button



**Step 11:** Paste the copied file



**Step 12:** Click on **OK** Button, then again OK button, again OK button.

Now, your JDK means java is installed for your system.

**Step 13:** To check JDK is installed in your system or not

- Open Command Prompt
- Type **java -version** command to check the version of JDK installed in your system

**JDK LINK FOR DOWNLOAD: - <http://cutt.ly/JDK64>**